CLASS SPORTS ROTATION SYSTEM

DSA Mini Activity — Circular Linked List

1. Objective

To develop a menu-driven C++ program using a circular linked list to store students' IDs and names, display the complete student list, and rotate the sports captain from one student to the next.

2. Problem Statement

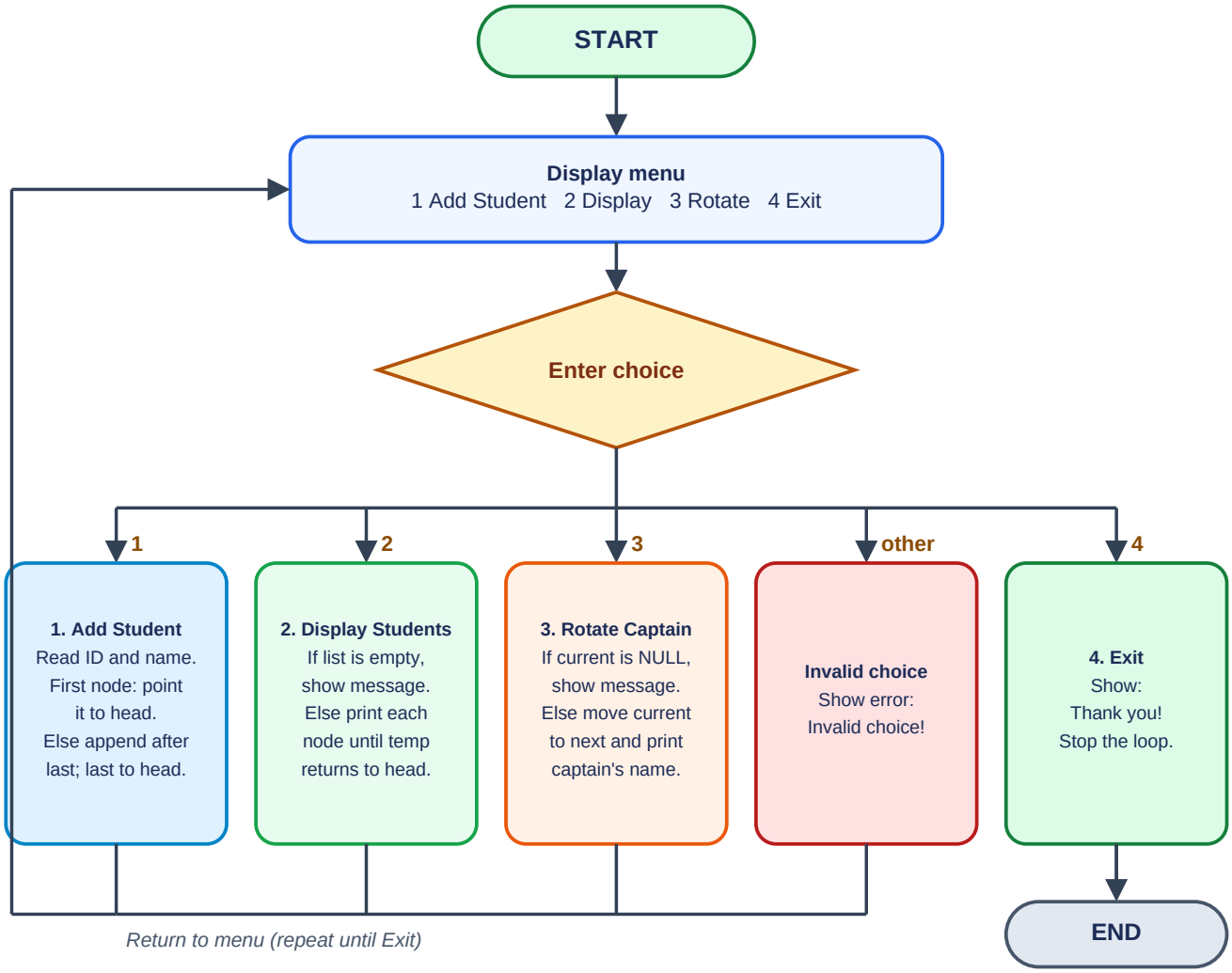
Create a Class Sports Rotation System in which students can be added dynamically. The program must display all students and allow the sports captain role to move to the next student each time the rotation option is selected. A circular linked list is used so that after the last student, rotation can continue again from the first student.

3. Algorithm

1. Start the program.
2. Initialize the pointers head and current to NULL.
3. Display the menu: Add Student, Display Students, Rotate Captain, and Exit.
4. Read the user's choice.
5. If the choice is Add Student:
    a. Create a new node and read the student ID and full name.
    b. If the list is empty, set head to the new node, point its next to head, and set current to head.
    c. Otherwise, find the last node, link it to the new node, and point the new node's next to head.
6. If the choice is Display Students:
    a. If the list is empty, display "No students available!".
    b. Otherwise, start at head, print each student's ID and name, and stop when the pointer returns to head.
7. If the choice is Rotate Captain:
    a. If current is NULL, display "No students available!".
    b. Otherwise, move current to current->next and display the current student's name.
8. If the choice is Exit, display "Thank you!" and terminate the loop.
9. If the choice is invalid, display "Invalid choice!".
10. Repeat the menu until the user selects Exit.
11. Stop.

4. Flowchart

The flowchart below shows the main menu and the logic of the three operations.



Circular linked list rule: the next pointer of the last node always points back to the first node (head).

5. Applications

- Sports captain rotation in a class or college team.
- Assigning students turns for sports equipment, duties, or team leadership.
- Round-robin scheduling, where each participant gets a turn in sequence.
- Circular queues and cyclic task scheduling in computer systems.
- Managing repeated rotations in clubs, classroom activities, and group tasks.

Conclusion

The Class Sports Rotation System demonstrates how a circular linked list can store student records and support continuous rotation. Because the last node points back to the head node, the sequence can repeat without reaching a NULL link during normal rotation.